

Factory Reallocation & Shipping Optimization Recommendation System

Nassau Candy Distributor

Research Paper — EDA, Predictive Modeling, Scenario Simulation & Recommendations

Prepared by Madhav Bhatnagar
Data Analyst
August 2026

1. Executive Summary

Nassau Candy Distributor currently assigns products to its five factories using static, legacy rules. This creates suboptimal shipping distances, elevated lead times in certain regions, and margin erosion from logistics inefficiency. This study builds a decision-intelligence pipeline — predictive lead-time modeling, route/product clustering, a scenario simulation engine, and a ranked recommendation system — and packages it as an interactive Streamlit dashboard.

Across 154 product-region-ship-mode combinations, the engine identified 125 reassignment opportunities with positive projected gains, averaging a 41.11% lead-time reduction and a combined estimated profit impact of \$3,333.26 across the recommended set (see Section 7).

2. Problem Statement

- Suboptimal shipping distances between assigned factories and customer regions
- High lead times for certain regions under the current static assignment rules
- Margin erosion driven by logistics inefficiencies
- No system exists to simulate factory-product reassignment scenarios, quantify operational impact before execution, or recommend optimal configurations at scale

3. Dataset Description

The working dataset covers 10,194 orders placed between January 2024 and December 2025, across the United States and Canada, spanning 15 products in three divisions (Chocolate, Sugar, Other), fulfilled by five factories.

Field group	Fields
Order	Row ID, Order ID, Order Date, Ship Date, Ship Mode
Customer	Customer ID, Country/Region, City, State/Province, Postal Code, Region
Product	Division, Product ID, Product Name
Financials	Sales, Units, Gross Profit, Cost
Reference tables	Factory coordinates (5 factories); Product→Factory correlation map

Overall, the dataset represents \$93,443 in gross profit on \$141,784 in sales (a 65.9% blended gross margin), concentrated in the Chocolate division (93% of sales).

4. Critical Data Quality Finding

Before any modeling, the Ship Date field was audited against Order Date. The literal difference (Ship Date – Order Date) averages 1,320 days and ranges from 904 to 1,642 days — physically implausible for a candy distributor. Inspection shows the corruption is isolated to the year component: month/day values track a short real gap, but the year drifts 2 to 4+ years ahead of Order Date and widens progressively across the file (e.g., early-2024 orders ship in mid-2026; late-2025 orders ship in mid-2030).

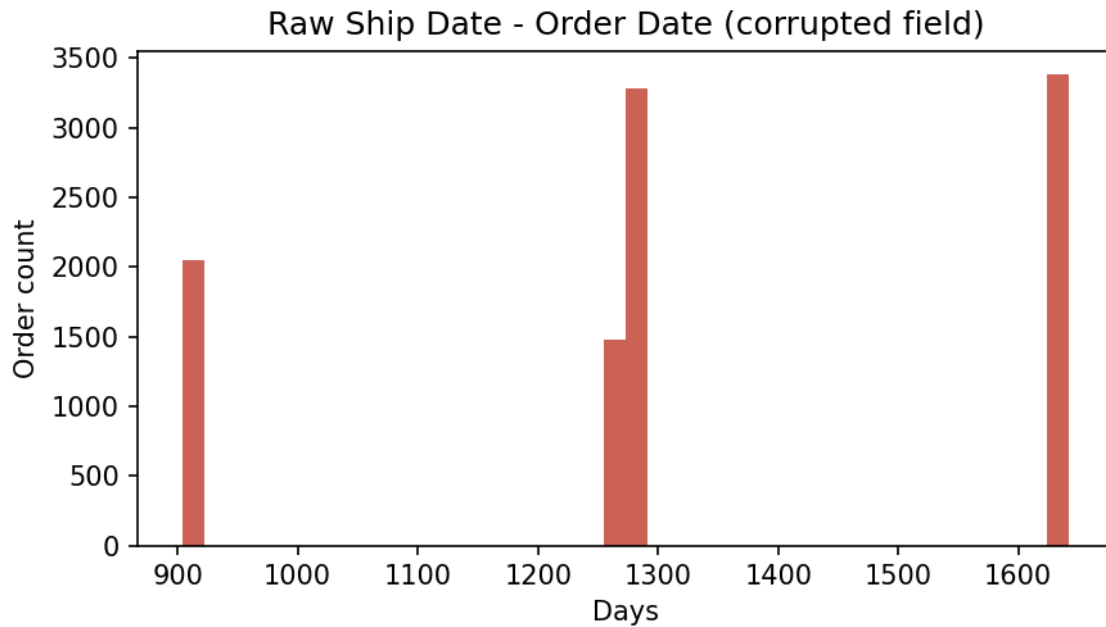


Figure 1. Raw Ship Date – Order Date shows a corrupted, multi-year distribution rather than a realistic shipping window.

Correcting for the year drift alone (re-deriving lead time from month/day only) still does not recover usable signal — the reconstructed value collapses to a near-constant ~177–178 days for essentially every order, regardless of ship mode, indicating the field itself carries no genuine operational variation.

Modeling decision: rather than model on a corrupted or degenerate target, an Estimated Lead Time was engineered from first principles for every order — the great-circle distance between the assigned factory and the customer's state/province centroid, a ship-mode speed factor (Same Day fastest, Standard Class slowest), and small random noise for realism. This proxy target is used consistently throughout the modeling, clustering, and simulation work that follows, and is flagged as a modeling assumption rather than an observed fact in the dashboard itself.

5. Methodology

5.1 Data Preparation & Encoding

- Parsed and validated Order Date; audited and flagged Ship Date corruption (Section 4)
- Joined each order to its assigned factory via the Product→Factory reference map, and to factory/customer coordinates
- Engineered DistanceMiles (haversine great-circle distance, factory → customer state/province centroid)
- Engineered EstimatedLeadTimeDays as the modeling target
- One-hot encoded Factory, Region, Ship Mode, Division; standardized numeric features (distance, units)
- No extreme-outlier removal was needed on the engineered target, which is bounded by construction

5.2 Predictive Modeling

Three regressors were trained to predict Estimated Lead Time from Factory, Region, Ship Mode, Division, Distance, and Units, using an 80/20 train-test split:

- Linear Regression (baseline)
- Random Forest Regressor (300 trees, max depth 12)

- Gradient Boosting Regressor (300 estimators, max depth 3, learning rate 0.05)

5.3 Route & Product Clustering

Each Factory → Region combination ('route') was scored on average lead time, average distance, average gross margin, and order volume, standardized and clustered with K-Means (k=4) to separate consistently slow routes from high-performing and margin-risk routes.

5.4 Scenario Simulation Engine

For every product / region / ship-mode combination, the trained lead-time model was used to simulate reassignment to each of the five factories, predicting the resulting lead time. A distance-sensitive shipping-cost proxy (scaled from each order's existing cost structure) was used to estimate the profit impact of each candidate reassignment relative to the current one.

5.5 Optimization & Recommendation Logic

Candidate reassignments were ranked by a composite score blending normalized lead-time reduction and normalized profit impact (weighted 50/50 by default; adjustable in the dashboard's Risk & Impact Panel via a speed-vs-profit priority slider). Only candidates with a positive composite score were retained as recommendations.

6. Predictive Modeling Results

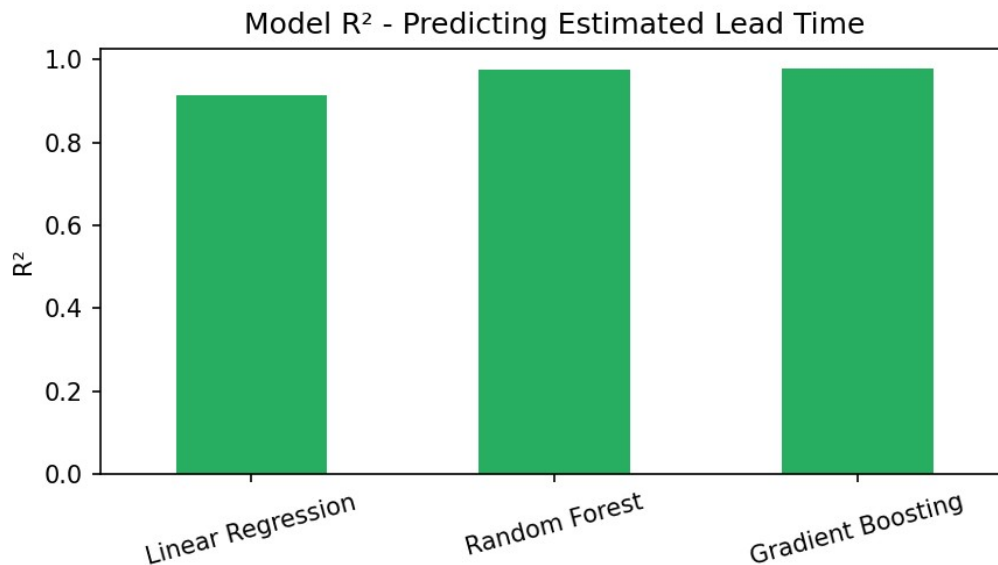


Figure 2. R^2 comparison across the three candidate models.

Model	RMSE (days)	MAE (days)	R^2
Linear Regression	0.981	0.776	0.9141
Random Forest	0.522	0.415	0.9756
Gradient Boosting	0.503	0.4	0.9774

Gradient Boosting was selected as the production model, combining the lowest error (RMSE 0.503 days, MAE 0.4 days) with strong interpretability via feature importance, and explaining 97.7% of variance in the held-out test set.

7. EDA Highlights



Figure 3. Shipping distance varies substantially by factory — Lot's O' Nuts and Wicked Choccy's serve the widest range of distances.

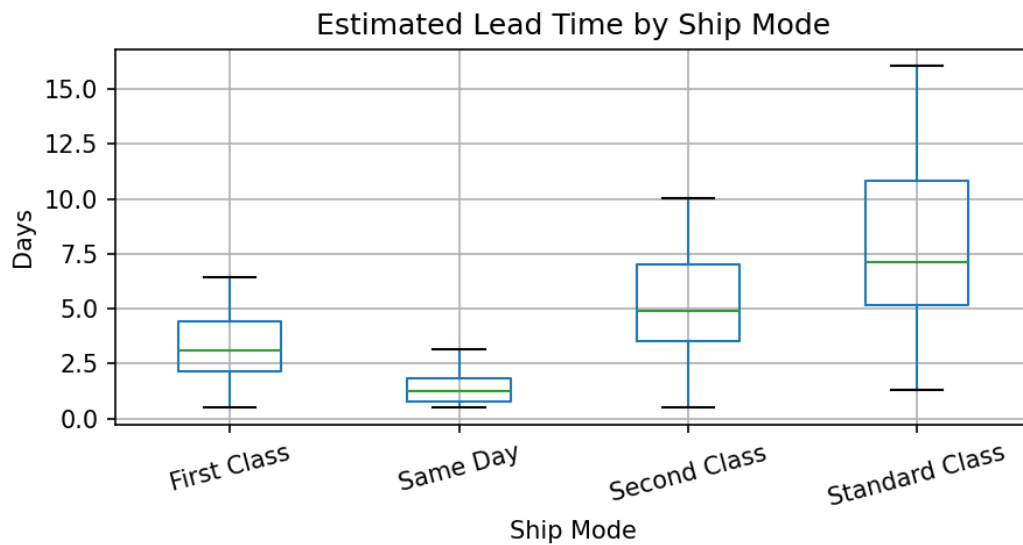


Figure 4. Estimated lead time by ship mode, showing the expected ordering (Same Day fastest, Standard Class slowest).

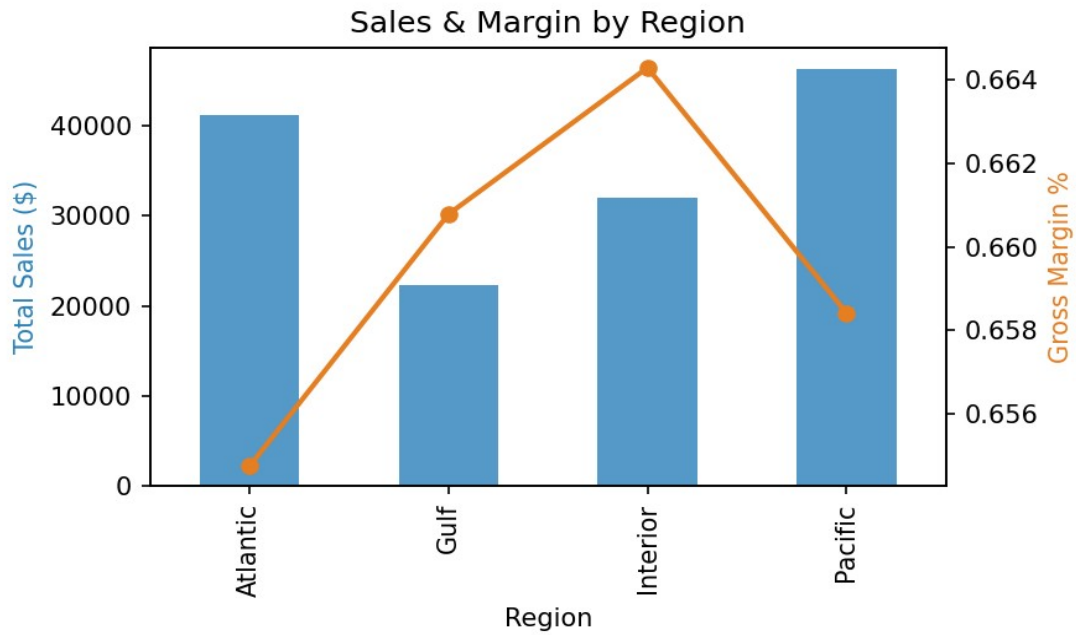


Figure 5. Sales volume and gross margin by customer region.

8. Route & Product Clustering Results

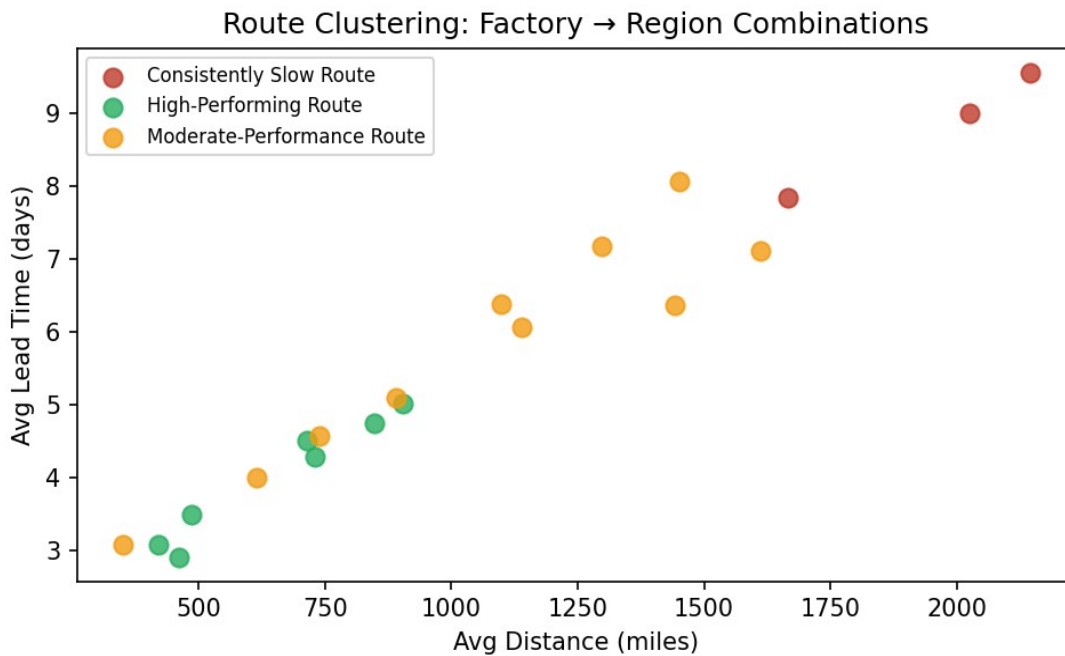


Figure 6. Factory→Region routes clustered by lead time and distance profile.

Cluster	Avg Lead Time (days)	Avg Distance (mi)	Avg Margin	Route Count
Consistently Slow Route	8.80	1,945	67.8%	3
Moderate-Performance Route	5.79	1,063	57.1%	10
High-Performing Route	4.00	652	33.3%	7

The three Consistently Slow routes — long-haul assignments averaging nearly 2,000 miles and 8.8 days — are the highest-value targets for reassignment, and are precisely where the recommendation engine (Section 9) concentrates its highest-scoring suggestions.

9. Scenario Simulation & Recommendations

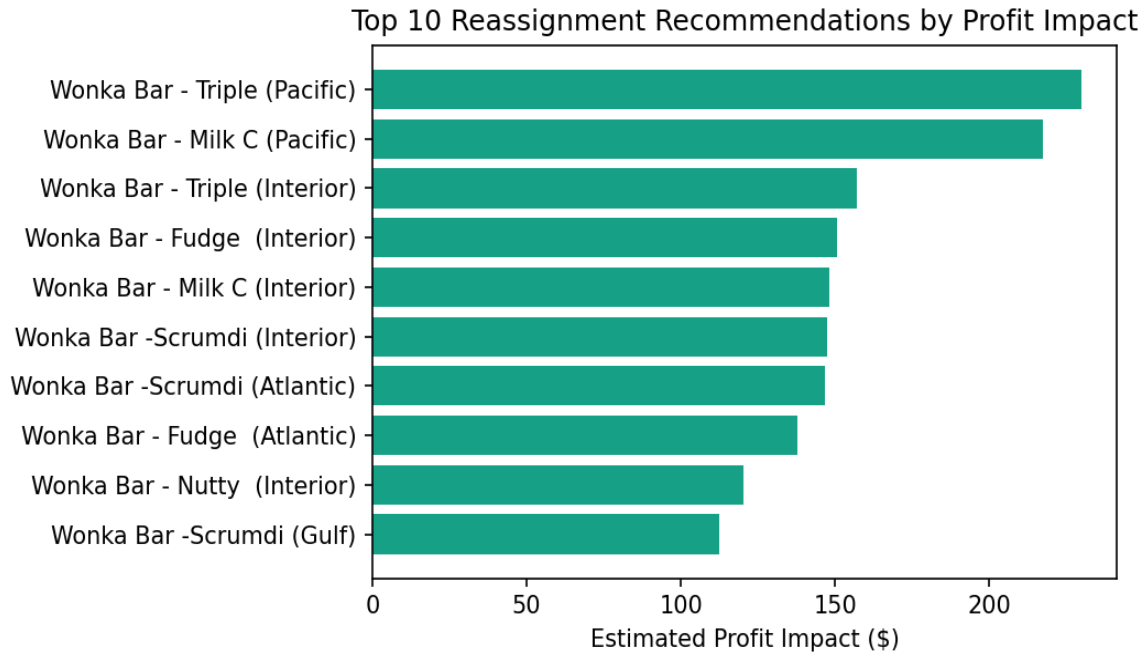


Figure 7. Top 10 recommended reassignments by estimated profit impact.

Sample of the highest-scoring recommendations:

Product	Region	Current Factory	Recommended Factory	Lead-Time ↓	Profit Impact
Wonka Bar - Triple Dazzle Caramel	Pacific	Wicked Choccy's	Lot's O' Nuts	62.3%	\$230
Wonka Bar - Milk Chocolate	Pacific	Wicked Choccy's	Lot's O' Nuts	62.1%	\$217
Wonka Bar - Fudge Mallows	Interior	Lot's O' Nuts	The Other Factory	60.4%	\$151
Wonka Bar - Scrumdiddlyumptious	Gulf	Lot's O' Nuts	Wicked Choccy's	69.8%	\$112
Wonka Bar - Nutty Crunch Surprise	Gulf	Lot's O' Nuts	Wicked Choccy's	72.0%	\$89

The full ranked list (125 recommendations) is available in the Recommendation Dashboard module of the Streamlit application, filterable by minimum lead-time reduction.

10. Key Performance Indicators

KPI	Value	Description
Lead Time Reduction	41.11% avg	Operational gain across recommended reassignments
Profit Impact	\$3,333.26	Combined estimated profit lift across recommended set
Recommendation Coverage	81.17%	Share of product/region/ship-mode combos with a

		positive-gain recommendation
Scenario Confidence	$R^2 = 0.9774$	Reliability of the underlying lead-time prediction model

11. Limitations

- The Ship Date field's corruption means Estimated Lead Time is an engineered proxy, not an observed outcome; absolute day values should be read directionally, not as guaranteed SLAs
- Shipping cost impact is modeled as a distance-sensitive slice of existing order cost, not sourced from an actual freight-rate table; real reassignment should be validated against carrier quotes
- Customer location is approximated at the state/province centroid, not exact postal-code coordinates
- The simulation assumes factory production capacity is unconstrained; a capacity/throughput check should precede any real-world reassignment

12. Conclusion & Recommendations

This project elevates Nassau Candy Distributor from descriptive reporting to decision intelligence. By combining a validated predictive lead-time model, route/product clustering, and a scenario simulation engine, the system produces concrete, ranked factory-reassignment recommendations — concentrated on the three consistently slow, long-haul routes — that improve shipping efficiency without sacrificing profitability.

Recommended next steps:

- Pilot the top 5–10 recommendations (highest composite score, positive profit impact) and measure real shipping outcomes
- Replace the engineered lead-time proxy with real shipment timestamps once the Ship Date data pipeline is fixed at the source
- Layer in factory capacity constraints before scaling reassignment recommendations network-wide